Sample Preparation

After sample received: 4ul of liquid sample was resuspend in 30ul of 100mMTris, pH8.5 with 8M Urea. TCEP and CAM treated for 30mins in dark, then dilute Urea to 2M by Tris and add 5ul of 0.1 ug/ul Trypsin/LysC (Promega # V5071) on each sample at 37 degrees overnight. Adding 7ul of FA to stop reaction. Aliquot 2x 25ul of reaction on two EvoTips for DIA and DDA on Flex.

LC-MS Analysis of Peptides

UHPLC

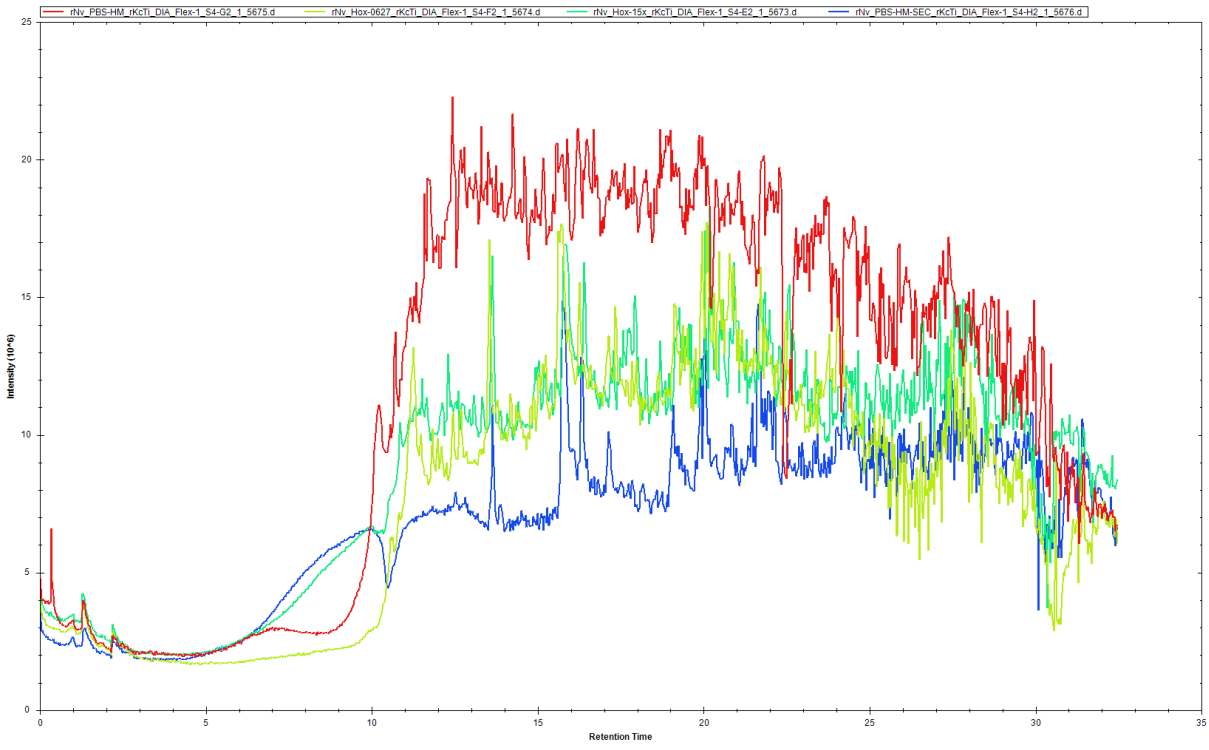
Peptides were resolved using an Evosep One EV1000 UHPLC in Zoom mode connected to an Aurora Elite CSI 15 cm x 75 µm C18 UHPLC column (ionopticks) in turn connected to a CaptiveSpray source. The Whisper Zoom 40SPD chromatography method with a runtime of 32.5 minutes was used with Buffer A (0.1% formic acid in LC-MS grade water) and Buffer B (0.1% formic acid in LC-MS grade acetonitrile).

Mass Spectrometry

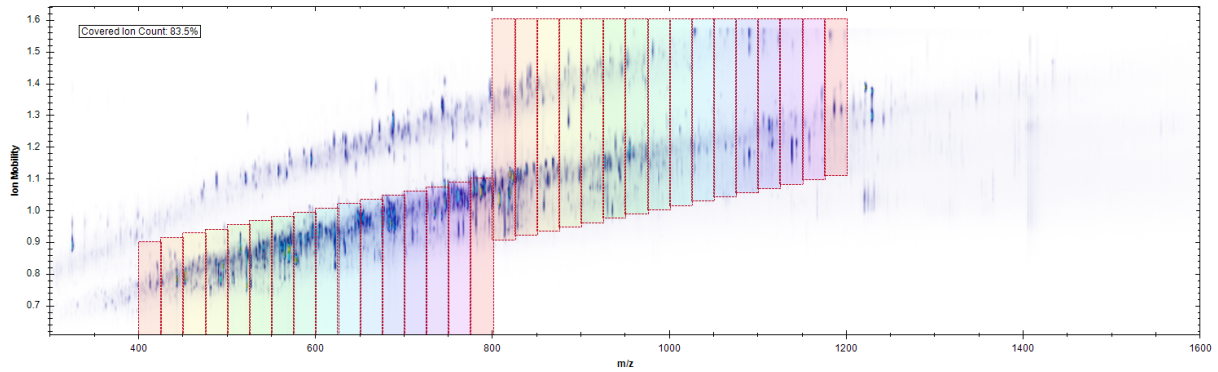
DIA and DDA mass spectrometry analysis used a timsTOF Flex MALDI 2 instrument run in positive ion mode. DIA was performed with MS1 data collected in the range 100-1700 m/z and in the range1.30-0.85 Vs/cm^2 for 1/K0. The TIMS ramp time was 100 ms. DIA used 32 windows (26 m/z wide) with data collected using 16 PASEF scans per cycle. Collision energies varied linearly with 1/K0 from 20.00 eV (1/K0 = 0.60 Vs/cm^2) to 59.00 eV (1/K0 = 1.60 Vs/cm^2).

DIA Method was used dia-PASEF - long gradient_1400V.m.

DIA data: Total Ion Chromatogram (TIC) for all samples

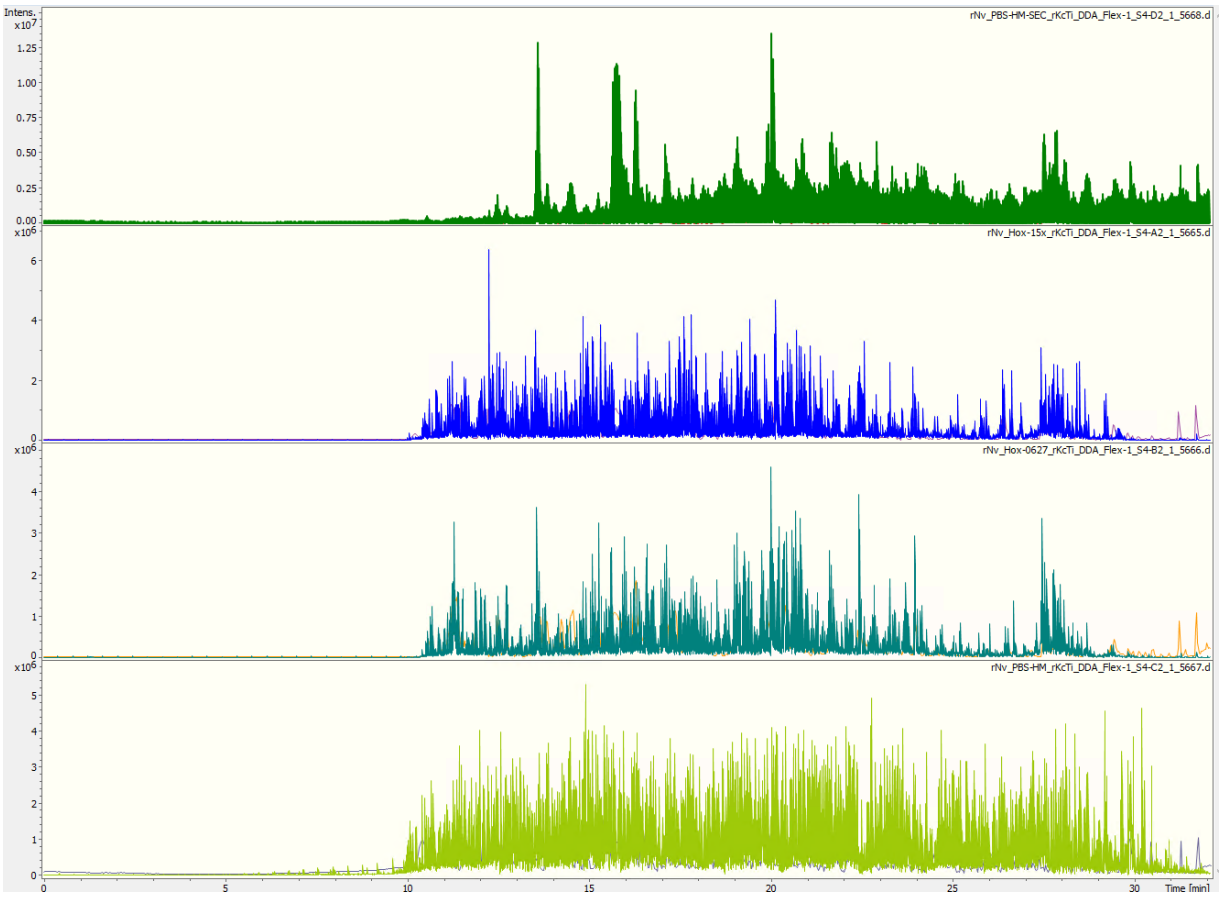


Ion mobility overview (DIA windows)



DDA Method was used DDA-PASEF-lowamount_1400V-1p9sec.m

DDA data: Total Ion Chromatogram (TIC) for all samples



Data Analysis

DIA Data Analysis with Spectronaut, DDA Data Analysis with FragPipe

Data was searched against the fasta database as below:

DIA: E_coli_UniProt_UP000002032_469008_NR_wCont_2025-04-01_PBX-HOX-MEISHM.fasta

DDA: 2025-09-29-decoys-E_coli_UniProt_UP000002032_469008_NR_wCont_2025-04-01_PBX-HOX-MEISHM.fasta

This database contains *Ecoli.* proteins downloaded from UniProt on 2025 and 381 contaminant proteins as published by Frankenfield et al. in 2022[3]. Also include three recombinational sequences that are shared by CPR 3 Sequences.

DIA Search method in Spectronaut was changed from default to "MS-DAP_ Pepline_ directDIA_ stringent". Variable Modifications set as Acetyl (Protein N-term) and Oxidation (M).

DDA Search method in FragPipe chose Basic-Search as Workflow and keep rest as default.

Results

Mass spectrometry Overview

Following the initial searches, we identified the following in each sample from DIA and DDA:

	DDA				DIA				
	Hox_ 0627	Hox_ 15x	PBS_H M	PBS_ HM_ SEC	Hox_ 0627	Hox_ 15x	PBS_HM	PBS_HM _ SEC	
Total Proteins	296	185	869	159	428	209	1338	184	**Total Proteins**
Total Peptides	1644	1503	4779	613	2312	1596	12066	848	**Total Peptides**
Total PSM	5892	6847	14080	4173	2841	2023	15737	1095	**Total precursors**

Protein Identification & Quantitation

The **bait protein, Hox1-His, MEISHM-His and PBX-His** were on the top of the table based on their **Combined Total Spectral Count** values in DDA-Pr and **NrOfPrecursorsIdentified** in DIA-Pr sheet. And a bunch of RPL proteins on the list. Trypsin and Protease IV are the proteins on the top of contaminant which are added in. Protein contaminants such as keratins or enzymes are listed as well. We use the ratio of Spectral Count(DDA) or Precursor Count(DIA) from contaminant proteins over total protein PG.Quantity to estimate contamination levels. The contamination level is okay since major contaminants are Trypsin and rLys-C we added in.

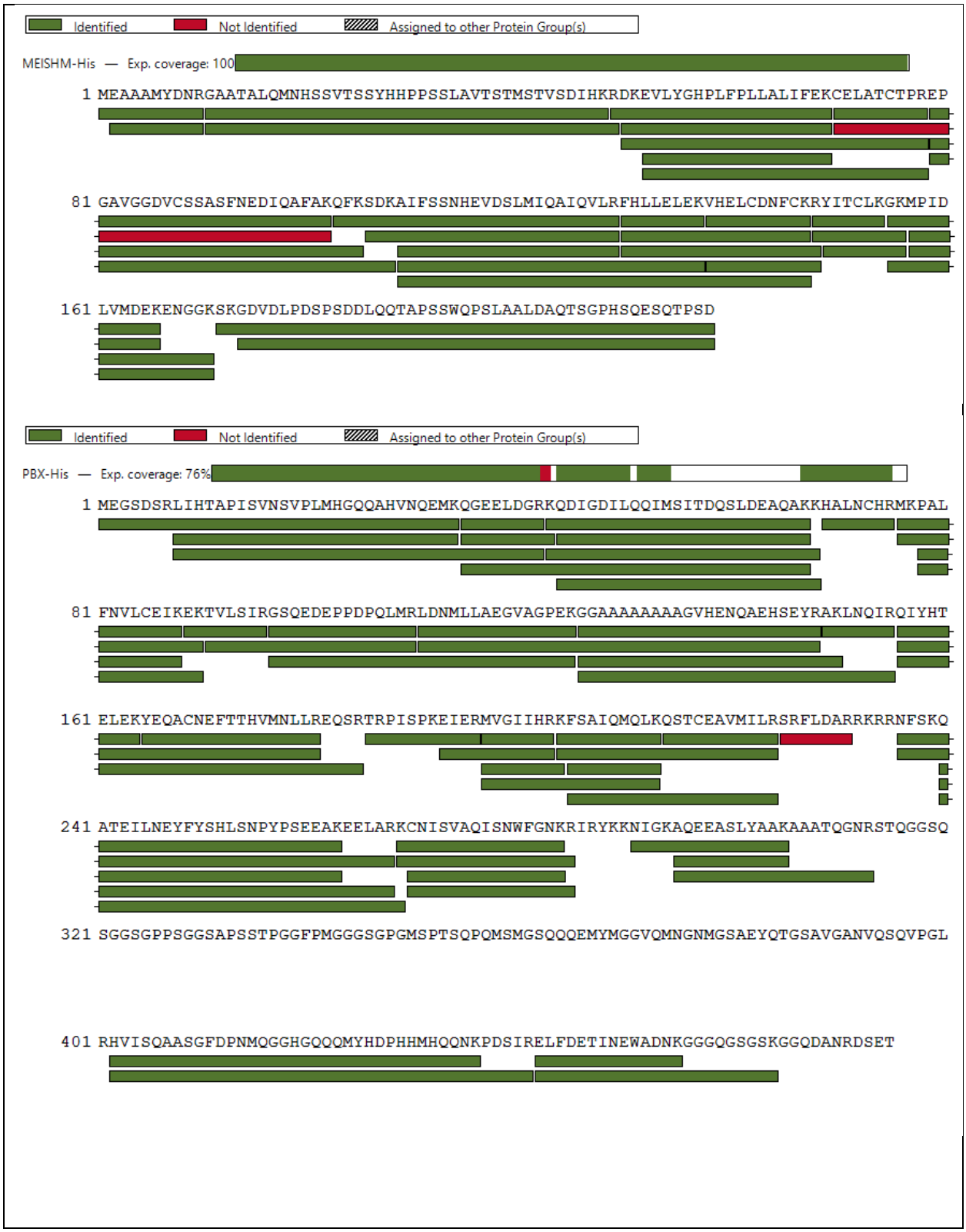
	DDA				DIA				
	Hox_0627	Hox_15x	PBS_HM	PBS_HM_SEC	Hox_0627	Hox_15x	PBS_HM	PBS_HM_SEC	
Total PSM	5892	6847	14080	4173	2841	2023	15737	1095	**Total precursor**
PSM for Hox1-His	959	669	275	189	44	39	14	10	**Precursor for Hox1-Hi**
	16.28%	9.77%	1.95%	4.53%	1.55%	1.93%	0.09%	0.91%	
PSM for MEISHM-His	0	0	1715	875	4	4	102	62	**Precursor for MEISHM-Hi**
	0.00%	0.00%	12.18%	20.97%	0.14%	0.20%	0.65%	5.66%	
PSM for PBX-His	0	0	2067	835	4	6	159	86	**Precursor for PBX-Hi**
	0.00%	0.00%	14.68%	20.01%	0.14%	0.30%	1.01%	7.85%	
PSM for Contaminant	2157	4853	903	1661	449	924	277	345	**Precursor for Contaminar**
	36.61%	70.88%	6.41%	39.80%	15.80%	45.67%	1.76%	31.51%	

The coverage data for **His-Hox1 (67%), MEISHM-His (100%) and PBX-His (76%)** can be found below for DIA. The DDA-FragPipe provides coverage maps for each protein of each sample which I saved in the DDA-coverage sheet. Hope it helps you!

Comparing the two data acquisition and analysis methods, I believe the DDA will be the winner to present your target proteins purity! I will run DDA on your purified samples in future!

Let me know if you have any questions!




Identified
Not Identified
Assigned to other Protein Group(s)
MEISHM-His — Exp. coverage: 100
1 MEAAAMYDNRGAATALQMNHSSVTSSYHHPPSSLAVTSTMSTVSDIHKRDKEVLYGHPLFPLLALIFEKCELATCTPREP
81 GAVGGDVCSSASFNEDIQAFAKQFKSDKAIFSSNHEVDSLMIQAIQVLRFHLLELEKVHELCDNFCKRYITCLKGKMPID
161 LVMDEKENGGKSKGDVDLPDSPSDDLQQTAPSSWQPSLAALDAQTSGPHSQESQTPSD
Identified
Not Identified
Assigned to other Protein Group(s)
PBX-His — Exp. coverage: 76%
1 MEGSDSRLIHTAPISVNSVPLMHGQQAHVNQEMKQGEELDGRKQDIGDILQQIMSITDQSLDEAQAKKHALNCHRMKPAL
81 FNVLCEIKEKTVLSIRGSQEDEPPDPQLMRLDNMLLAEGVAGPEKGGAAAAAAAAGVHENQAEHSEYRAKLNQIRQIYHT
161 ELEKYEQACNEFTTHVMNLLREQSRTRPISPKEIERMVGIIHRKFSAIQMQLKQSTCEAVMILRSRFLDARRKRRNFSKQ
241 ATEILNEYFYSHLSNPYPSEEAKEELARKCNISVAQISNWFGNKRIRYKKNIGKAQEEASLYAAKAAATQGNRSTQGGSQ
321 SGGSGPPSGGSAPSSTPGGFPMGGGSGPGMSPTSQPQMSMGSQQQEMYMGGVQMNGNMGSAEYQTGSAVGANVQSQVPGL
401 RHVISQAASGFDPNMQGGHGQQQMYHDPHHMHQQNKPDSIRELFDETINEWADNKGGGQGSGSKGGQDANRDSET